

AI and IoT-Enabled Smart Office System with Cloud-Based Energy Optimization

25-26J-034

Preliminary Progress Report

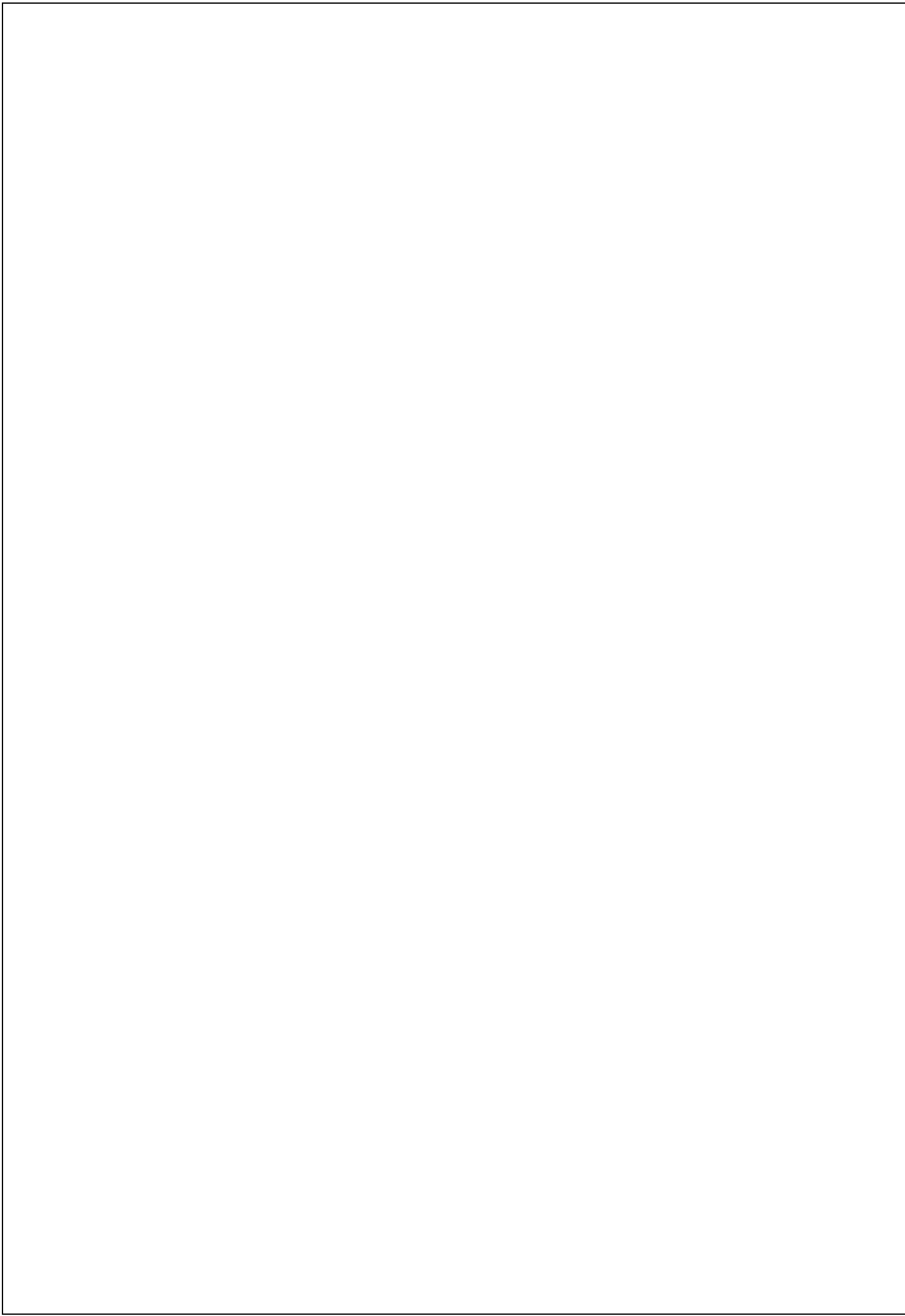
B.Sc. (Hons) in Information Technology Specializing in
Computer Systems and Network Engineering

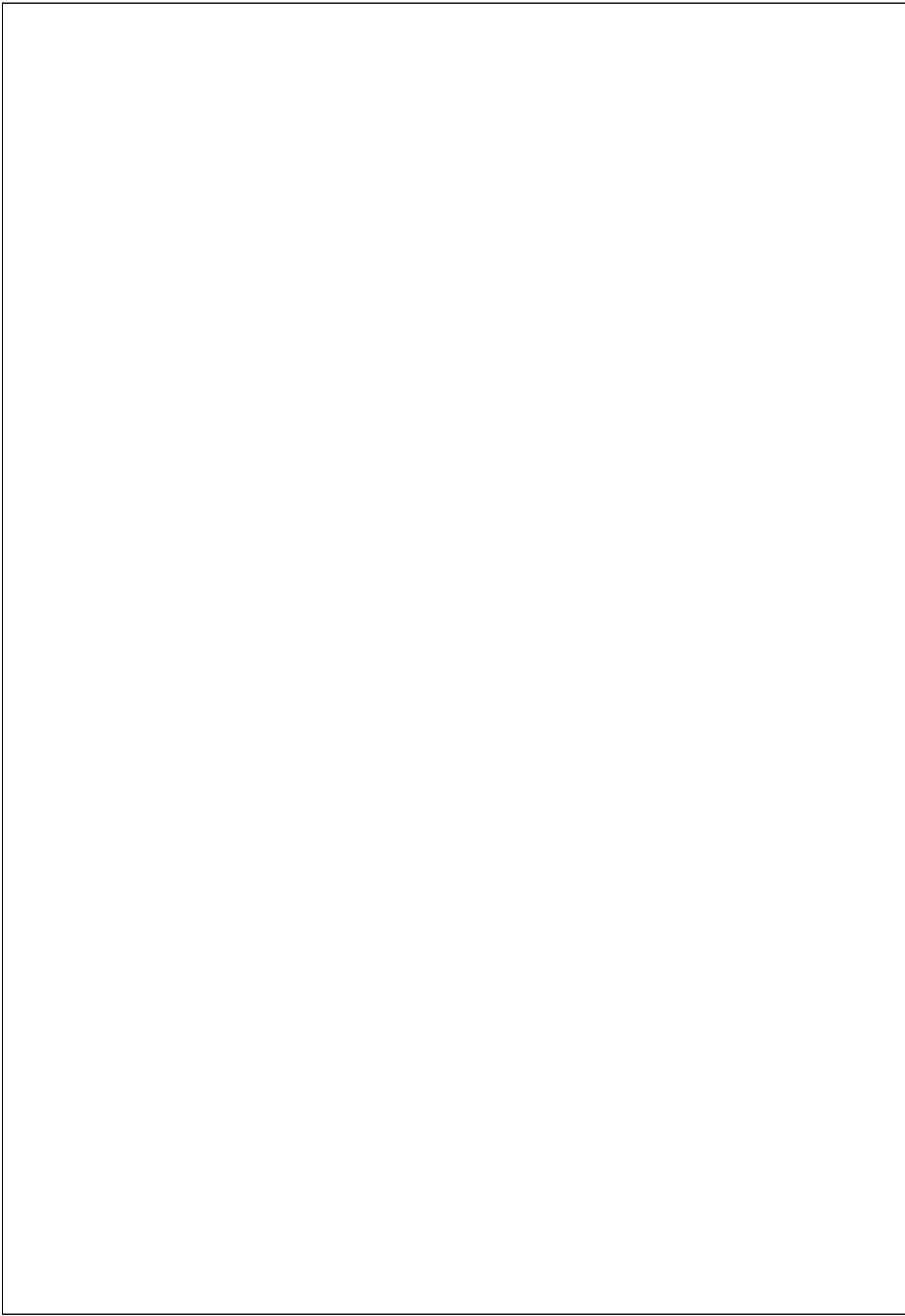
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Background

Modern office environments rely heavily on digital technologies such as video conferencing platforms, cloud-based software development tools, collaborative documentation systems, and data analytics applications. While these technologies improve productivity and operational efficiency, they also introduce new challenges related to employee stress, uneven workload distribution, and inefficient utilization of organizational resources. In many organizations, workloads are assigned manually and resources are managed using static rules, which fail to adapt to real-time changes in employee conditions and work demands.

Employee stress has become a significant concern in contemporary workplaces, as prolonged high stress levels can negatively impact performance, decision-making ability, and overall wellbeing. At the same time, poor workload management can lead to overburdening some employees while underutilizing others, reducing productivity and increasing burnout risks. Traditional management approaches lack intelligent mechanisms to continuously monitor stress levels, analyze workload patterns, and support data-driven decision making.

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Internet of Things (IoT) technologies provide opportunities to address these challenges. Machine learning models can analyze patterns in user activity, workload characteristics, and behavioral indicators to classify task types, estimate workload intensity, and predict system demands. Similarly, computer vision techniques enable non-intrusive stress detection using facial features captured through cameras. When combined with modern web technologies and scalable databases, these approaches can form intelligent systems capable of real-time monitoring and adaptive management.

This project proposes an AI-driven smart office stress and workload management system that integrates machine learning models, stress detection mechanisms, and interactive dashboards into a unified platform. The system is designed to monitor employee stress levels, classify and manage workloads, and support intelligent task assignment based on analytical insights. MongoDB is used for storing historical data, predictions, and system logs, enabling continuous analysis and future enhancements.

The project is implemented as a two-component group project, where one component focuses on machine learning-based workload analysis and optimization, and the other focuses on AI-based stress detection and workload management interfaces. At the current stage, baseline machine learning models and core system components have been developed and evaluated, establishing the foundation for further integration, automation, and optimization in subsequent phases.

Literature Analysis

Existing literature indicates that employee stress and workload imbalance are significant factors affecting productivity, decision-making quality, and overall organizational performance. Studies in occupational psychology and human–computer interaction emphasize that prolonged exposure to high workloads and unmanaged stress leads to burnout, reduced efficiency, and increased error rates. As a result, there has been growing interest in intelligent systems that can continuously monitor employee conditions and assist in adaptive workload management.

Recent research highlights the use of machine learning techniques for analyzing human activity patterns, task characteristics, and performance indicators in workplace environments. Classification models such as Logistic Regression, Random Forests, and Neural Networks have been widely applied to categorize task types and workload intensity based on behavioral and activity-related features. Regression-based models are also commonly used to predict resource usage trends, including bandwidth consumption and system load, enabling proactive optimization rather than reactive control.

In the area of stress detection, several studies demonstrate the effectiveness of computer vision and facial analysis techniques for non-intrusive stress assessment. Features extracted from facial expressions, eye movement, and muscle tension using tools such as OpenCV have been shown to correlate with stress-related indicators. Machine learning classifiers trained on these features can distinguish between different stress levels with reasonable accuracy under controlled conditions. However, the literature also notes challenges related to lighting conditions, facial occlusion, and individual variability, which necessitate robust feature extraction and model validation.

From a systems perspective, web-based dashboards and real-time visualization platforms are widely adopted to support monitoring and decision-making in intelligent management systems. Research emphasizes the importance of integrating analytics, historical data, and alert mechanisms into unified interfaces that are accessible to administrators and end users. NoSQL databases such as MongoDB are frequently used in such systems due to their flexibility in handling heterogeneous data, including logs, predictions, timestamps, and user-generated records.

Furthermore, several studies highlight the value of feedback-driven learning and adaptive systems, where model predictions and system outcomes are continuously evaluated and used to refine future decisions. Feedback loops improve model robustness and allow intelligent systems to adapt to changing user behavior and workload patterns over time.

Overall, the literature supports the feasibility of integrating machine learning-based workload analysis, computer vision-based stress detection, real-time dashboards, and scalable data storage into a single intelligent framework. However, many existing

solutions focus on isolated aspects such as stress monitoring or task management without combining predictive analytics, feedback mechanisms, and workload optimization within a unified smart office system. This project addresses this gap by proposing and implementing an integrated architecture that combines stress detection, workload analysis, intelligent task management, and data-driven visualization to support adaptive decision-making in modern office environments

Research Component

Component 01

Research Component Name

Multi-Model Machine Learning Framework for Workload Analysis and Intelligent Resource Optimization in Smart Office Environments

Component Overview

This research component focuses on the design, development, and preliminary evaluation of a **four-model Machine Learning framework** to analyze office workload behavior and support intelligent resource optimization. The component aims to transform raw activity and workload-related data into actionable insights that can be used for workload balancing, system optimization, and decision support in smart office environments.

The component is entirely software-based at the current stage and serves as the **core analytical engine** of the overall system.

Key Features

1. Task Type Classification Using Machine Learning (Model 01)

- This model is responsible for identifying the **type of task** being performed by an employee.
- Supported task categories include:
 1. Video conferencing
 2. Software development
 3. Data analysis
 4. Large file transfer

5. Documentation and creative work

- Input features are derived from workload-related indicators and usage patterns extracted from the dataset.
- Multiple supervised learning algorithms were implemented and compared, including:
 1. Logistic Regression
 2. Random Forest
 3. Neural Networks
- This model acts as the **first decision layer** in the system, providing contextual understanding of user activity.
- The classified task type is passed as an input to subsequent models for deeper analysis.

2. Workload Intensity Classification (Model 02)

- This model estimates the **intensity of the workload** associated with each task.
- Workload levels are classified into:
 1. Low
 2. Medium
 3. High
- Input features include:
 - ✓ Output from the task classification model
 - ✓ Additional workload indicators such as frequency, duration, and usage patterns
- Random Forest and Neural Network models were primarily used due to their ability to capture complex, non-linear relationships.
- This model supports:
 - ✓ Identification of overworked employees
 - ✓ Workload balancing decisions
- It forms a **bridge between task understanding and resource optimization**.

3. Resource / Bandwidth Usage Prediction Using Regression (Model 03)

- This model predicts expected **resource demand** (e.g., network bandwidth or system load) based on:
 - ✓ Task type
 - ✓ Workload intensity
 - ✓ Historical usage patterns
- Implemented using regression-based machine learning techniques.
- The model enables **proactive planning** by estimating future resource requirements instead of reacting to congestion or overload.
- Although bandwidth enforcement is simulated at this stage, the model is designed to integrate with:
 - ✓ SD-WAN controllers
 - ✓ QoS policy engines
- This model supports intelligent decision-making for system optimization.

4. Feedback-Based Learning and Optimization Model (Model 04)

- This model introduces **adaptive intelligence** into the system.
- It uses feedback collected from:
 - ✓ Past predictions
 - ✓ System performance outcomes
 - ✓ Historical workload patterns
- The feedback model:
 - ✓ Identifies inefficiencies such as over-allocation or under-allocation
 - ✓ Supports incremental improvement of prediction accuracy
- This component enables a **closed-loop learning mechanism**, where the system continuously improves based on observed outcomes.
- It lays the foundation for long-term adaptability and robustness

Performance Evaluation Metrics

1. Task Type Classification Evaluation (Model 01)

The task classification model is a **multi-class classification problem**. The following metrics were used:

- **Accuracy**
Measures the overall proportion of correctly classified task instances across all task categories.
- **Precision**
Evaluates how many instances predicted as a specific task type are actually correct, indicating prediction reliability.
- **Recall**
Measures the ability of the model to correctly identify all instances of a given task type.
- **F1 Score (Macro and Weighted)**
Provides a balanced measure of precision and recall, particularly useful when task categories are not evenly distributed.
- **Confusion Matrix**
Displays class-wise prediction performance and helps identify misclassification patterns between task categories.
- **Area Under the Curve (AUC)**
Evaluates the model's ability to distinguish between task classes by analyzing prediction confidence.

2. Workload Intensity Classification Evaluation (Model 02)

The workload intensity model classifies workloads into **Low, Medium, and High** categories. The following metrics were applied:

- **Accuracy**
Measures the correctness of workload intensity predictions.
- **Precision and Recall**
Evaluate the reliability and completeness of predictions for each workload level.
- **F1 Score (Micro and Macro)**
Assesses balanced performance across workload categories.
- **Hamming Loss**
Measures the fraction of incorrectly predicted labels, providing insight into overall classification error.
- **Confusion Matrix**
Helps analyze misclassification between workload intensity levels.

3. Resource / Bandwidth Prediction Evaluation (Model 03)

The resource prediction model is a **regression-based model**. The following metrics were used:

- **Mean Squared Error (MSE)**
Measures the average squared difference between predicted and actual resource usage values.
- **Root Mean Squared Error (RMSE)**
Provides an interpretable error metric in the same unit as the predicted resource values.
- **Mean Absolute Error (MAE)**
Measures the average magnitude of prediction errors without considering direction.
- **Predicted vs Actual Trend Analysis**
Compares predicted resource usage patterns with actual observed values to assess trend accuracy.

4. Feedback-Based Learning Evaluation (Model 04)

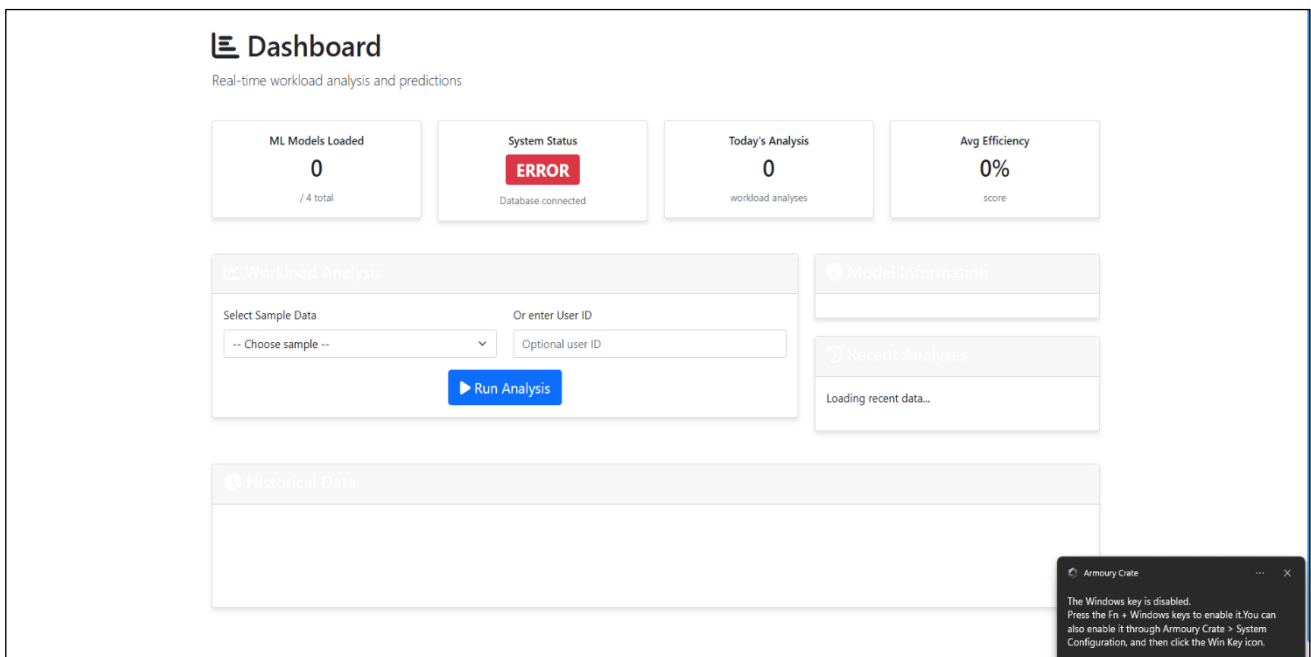
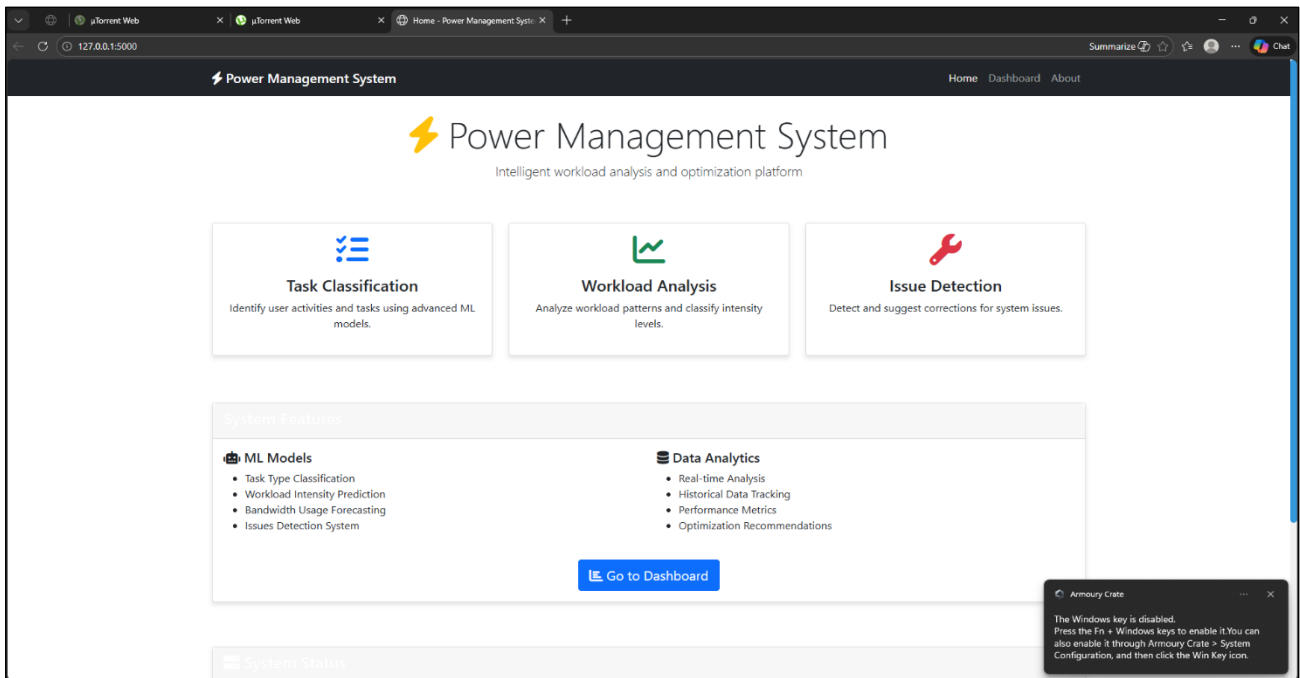
The feedback model evaluates system behavior over time rather than direct prediction accuracy. The following indicators were used:

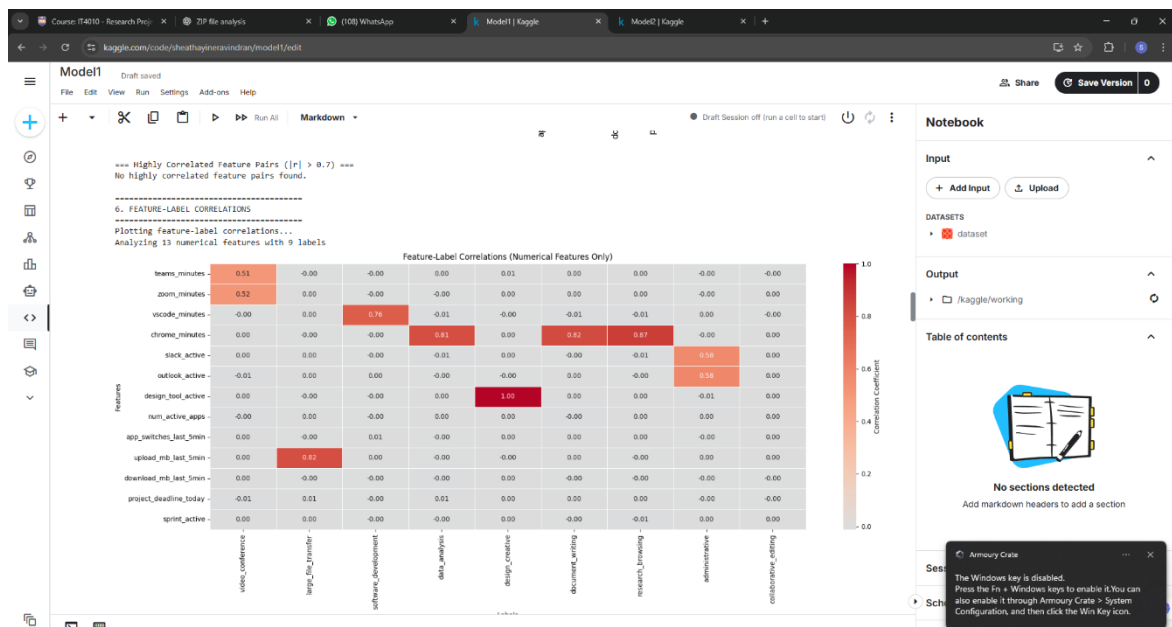
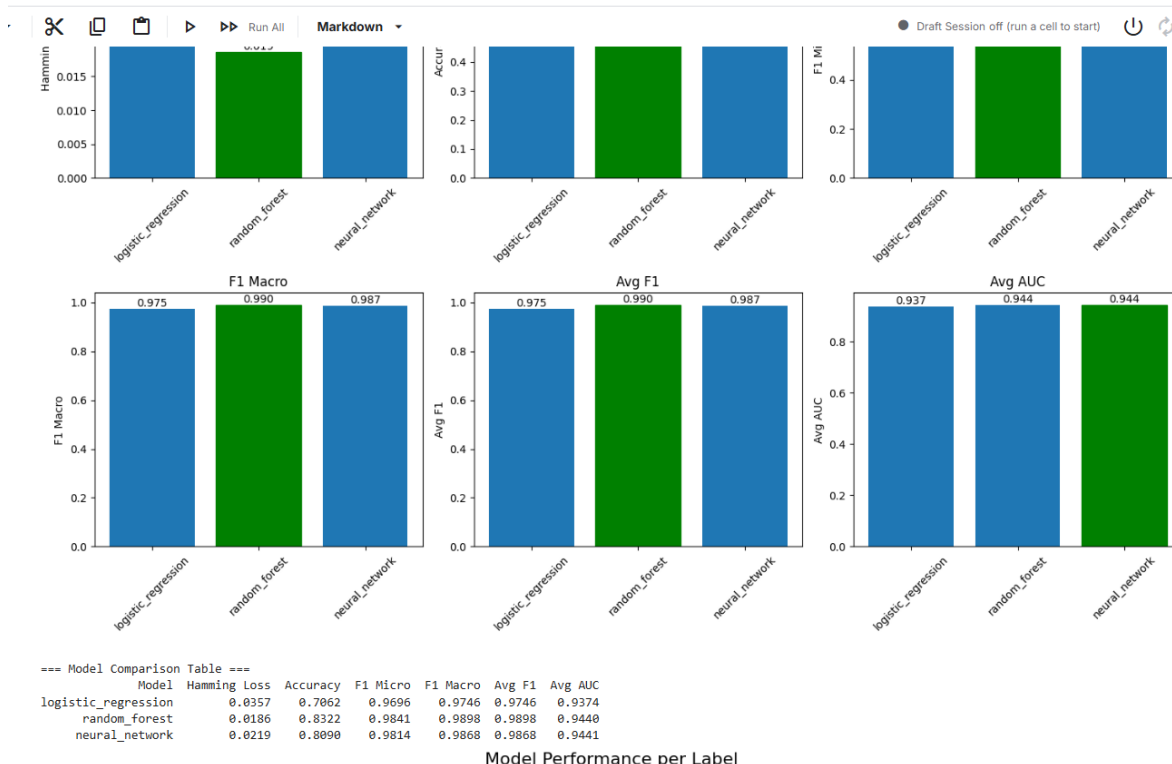
- **Prediction Error Reduction Over Time**
Measures improvement in prediction accuracy across multiple iterations.
- **Stability of System Decisions**
Evaluates consistency in resource allocation and workload decisions.
- **Historical Performance Comparison**
Compares earlier predictions with updated outcomes to assess learning effectiveness.

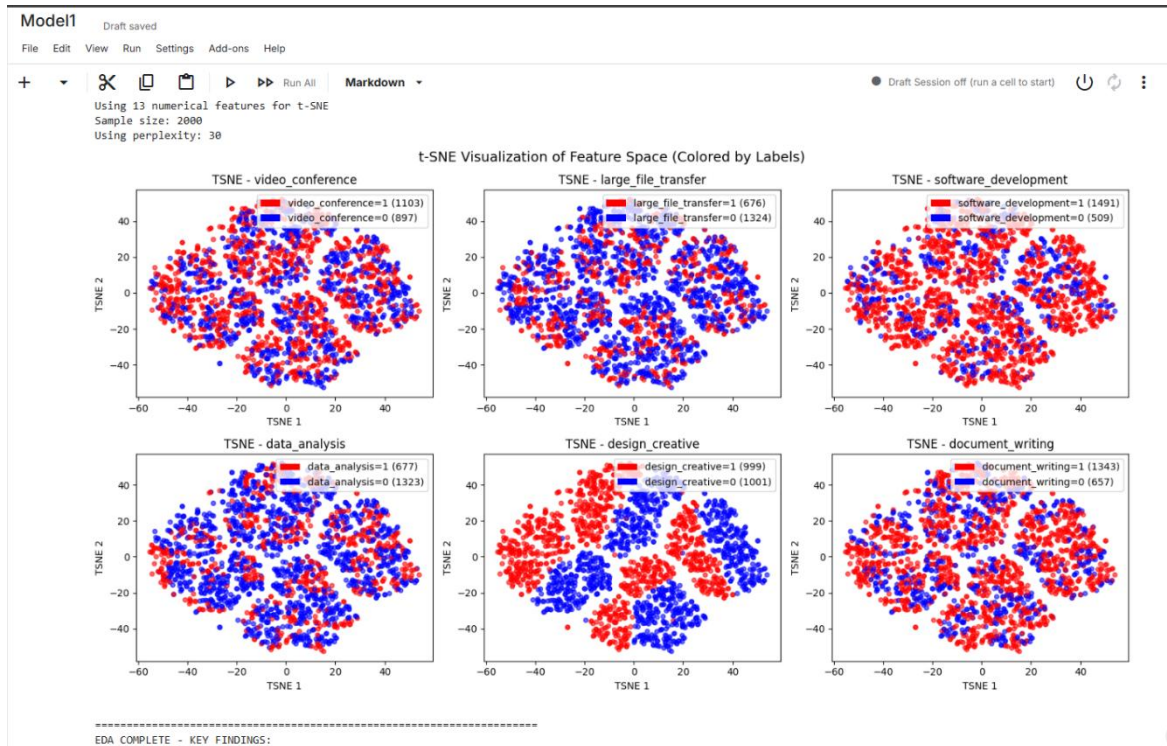
5. Feature Space and Data Quality Evaluation

To validate dataset quality and feature separability, the following techniques were applied:

- **t-SNE (t-Distributed Stochastic Neighbor Embedding)**
Used to visualize high-dimensional feature spaces and observe clustering behavior among task categories.
- **Cluster Separability Observation**
Visual inspection of clusters to confirm meaningful separation between different task and workload classes.







Model2 Draft saved

File Edit View Run Settings Add-ons Help

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13. download_mb_last_5min
14. project_deadline_today
15. sprint_active
16. priority_level
17. meeting_duration_minutes
18. unread_emails
19. unread_slack
20. workload_class
21. label_source
22. label_confidence

DATASET INFORMATION

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RangeIndex: 100000 entries, 0 to 99999
Data columns (total 22 columns):
Column Non-Null Count Dtype

0 user_id 100000 non-null object
1 timestamp 100000 non-null object
2 teams_minutes 100000 non-null int64
3 zoom_minutes 100000 non-null int64
4 vscode_minutes 100000 non-null int64
5 chrome_minutes 100000 non-null int64
6 slack_active 100000 non-null int64
7 outlook_active 100000 non-null int64
8 design_tool_active 100000 non-null int64
9 num_active_apps 100000 non-null int64
10 app_switches_last_5min 100000 non-null int64
11 upload_mb_last_5min 100000 non-null int64
12 download_mb_last_5min 100000 non-null int64
13 project_deadline_today 100000 non-null int64
14 sprint_active 100000 non-null int64
15 priority_level 100000 non-null object
16 meeting_duration_minutes 100000 non-null int64
17 unread_emails 100000 non-null int64
18 unread_slack 100000 non-null int64
19 workload_class 100000 non-null object
20 label_source 100000 non-null object
21 label_confidence 100000 non-null float64
dtypes: float64(1), int64(16), object(5)
memory usage: 16.8+ MB

Notebook

Input

+ Add Input Upload

DATASETS

dataset

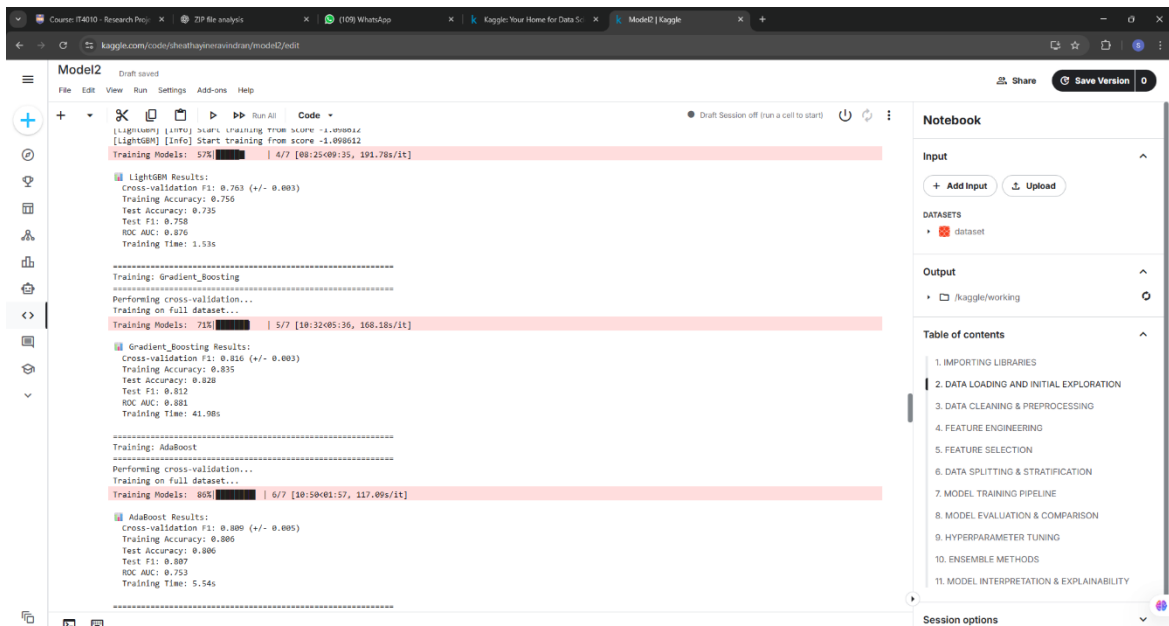
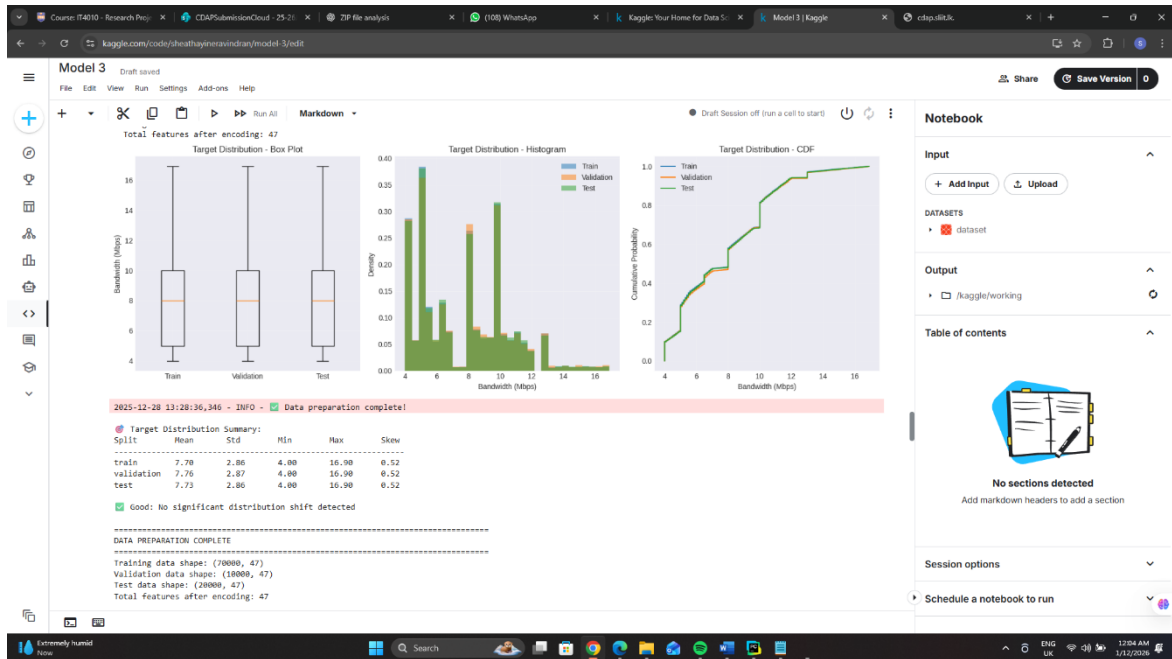
Output

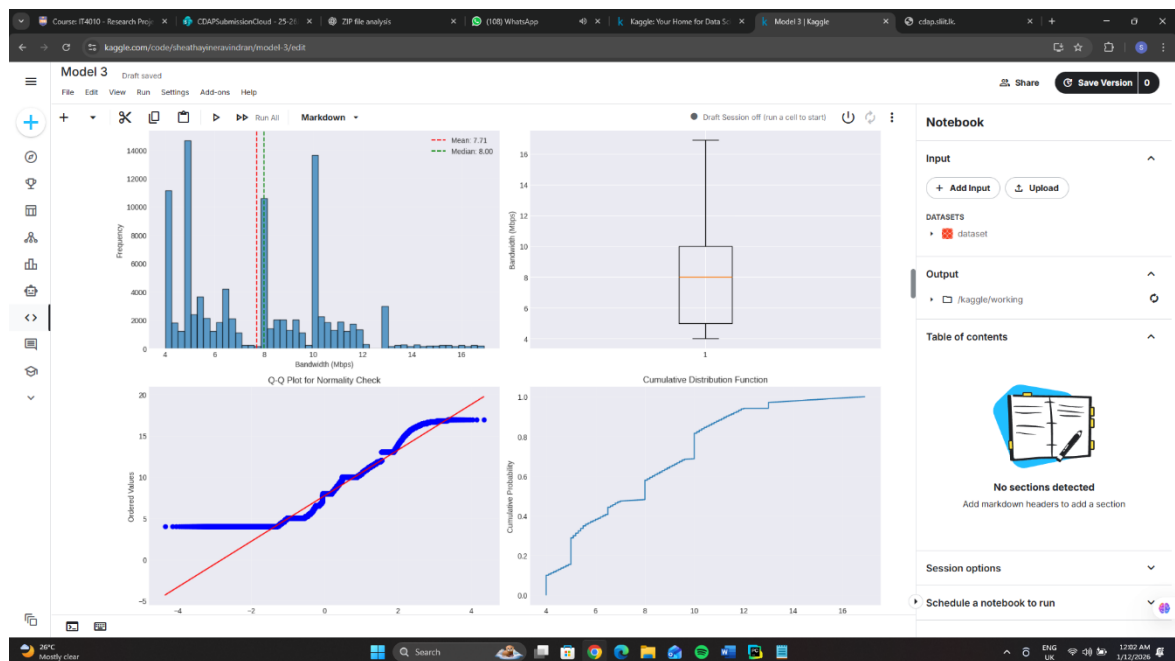
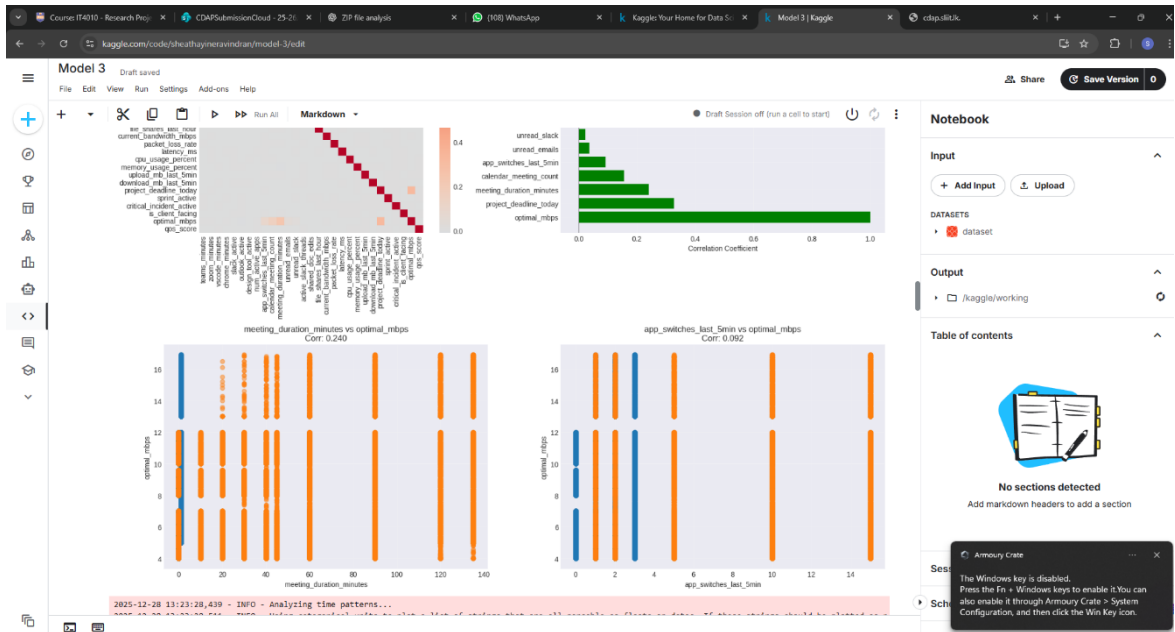
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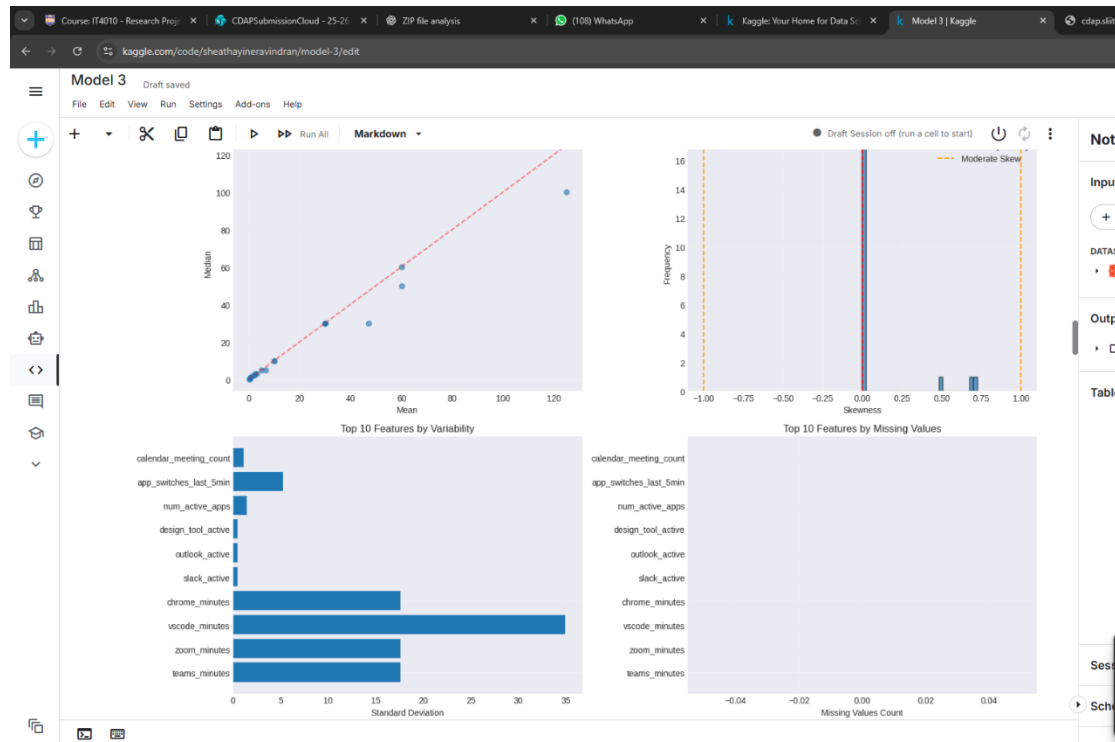
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1. IMPORTING LIBRARIES
2. DATA LOADING AND INITIAL EXPLORATION
3. DATA CLEANING & PREPROCESSING
4. FEATURE ENGINEERING
5. FEATURE SELECTION
6. DATA SPLITTING & STRATIFICATION
7. MODEL TRAINING PIPELINE
8. MODEL EVALUATION & COMPARISON
9. HYPERPARAMETER TUNING
10. ARMOURY CRATE
11. THE WINDOWS KEY IS DISABLED

The Windows key is disabled.
Press the Fn + Windows keys to enable it. You can also enable it through Armoury Crate > System Configuration, and then click the Win Key icon.







Results and Findings

1. Task and Workload Classification Results

- Machine learning models successfully classified task types and workload intensity with high baseline accuracy.
- Random Forest and Neural Network models demonstrated superior performance compared to baseline classifiers.
- Confusion matrix analysis shows balanced classification across workload categories.

2. Regression and Resource Prediction Results

- Regression-based model produced stable and consistent resource usage predictions.
- Predicted trends closely follow observed workload patterns.
- Results validate the feasibility of proactive resource optimization.

3. Overall System-Level Findings

- The multi-model approach enables **layered intelligence**, where each model contributes to a specific analytical function.
- Feature visualization confirms meaningful separation of task categories.
- The integrated four-model framework provides a strong foundation for intelligent workload management in smart office environments.
- Results obtained at this stage are considered **preliminary and baseline**, suitable for PP1 validation.

Current Limitations and Future Improvements

- Models are trained on controlled datasets; real-world deployment will require further validation.
- Hyperparameter optimization and extended datasets are planned.
- Integration with real network controllers and live monitoring systems is planned for future phases.

Component 02

Research Component Name

AI-Based Stress Detection and Intelligent Task Management Interface for Smart Office Environments

Component Overview

This research component focuses on the **human-centric intelligence and interaction layer** of the smart office system. It addresses the challenges of employee stress monitoring, workload visibility, and task management through the use of **AI-based stress detection, structured workload handling, and role-based dashboards**.

The component integrates stress analysis outputs with task and workload management features, enabling administrators to make informed decisions while allowing employees to monitor their own stress status and assigned tasks. The work is aligned with the **Software**

Requirements Specification (SRS) and emphasizes usability, clarity, and controlled access to sensitive information.

Key Features

1. AI-Based Stress Detection Module

- Implements a stress detection mechanism using **camera-based input**.
- Uses computer vision techniques to extract facial features associated with stress indicators.
- A machine learning classifier processes extracted features to predict stress levels.
- Stress levels are categorized into meaningful states (e.g., low, moderate, high).
- The module enables **non-intrusive and real-time stress monitoring** within office environments.
- Designed to support periodic stress checks triggered by the system or administrators.

2. Employee Stress Monitoring and History Tracking

- Allows employees to view:
 - ✓ Their current stress level
 - ✓ Historical stress trends over time
- Stress records are stored with timestamps, enabling trend analysis.
- Helps employees gain awareness of their workload and stress patterns.
- Supports transparency while maintaining privacy controls.

3. Intelligent Task and Workload Management Interface

- Provides functionality for administrators to:
 - ✓ Create workloads and tasks
 - ✓ Define workload priority and intensity levels
 - ✓ Assign tasks manually or through system logic
- Workloads are categorized (Low / Medium / High) to support structured distribution.
- Employees can view assigned tasks and update task status.
- The interface supports workload balancing based on stress indicators and workload rules.

4. Role-Based Dashboards (Admin & Employee)

- **Admin Dashboard**
 - ✓ Overview of employee stress levels
 - ✓ Workload distribution across users
 - ✓ Task management and monitoring tools
- **Employee Dashboard**
 - ✓ Personal stress status and history
 - ✓ Assigned workloads and task progress
- Role-Based Access Control (RBAC) ensures:
 - ✓ Employees can only view their own stress data
 - ✓ Admins have controlled visibility for management purposes

Results and Findings

1. Stress Detection Results

- The AI-based stress detection module successfully identifies stress levels using camera input.
- Preliminary results demonstrate consistent classification under controlled conditions.
- The approach validates the feasibility of using computer vision for workplace stress monitoring.
- Results are considered **baseline and preliminary**, suitable for PP1 evaluation.

2. Task and Workload Management Outcomes

- The workload management interface supports effective task creation, categorization, and assignment.
- Workload intensity labels improve clarity in task distribution.
- The integration of stress indicators with task management enables informed workload decisions.

3. Dashboard and Usability Findings

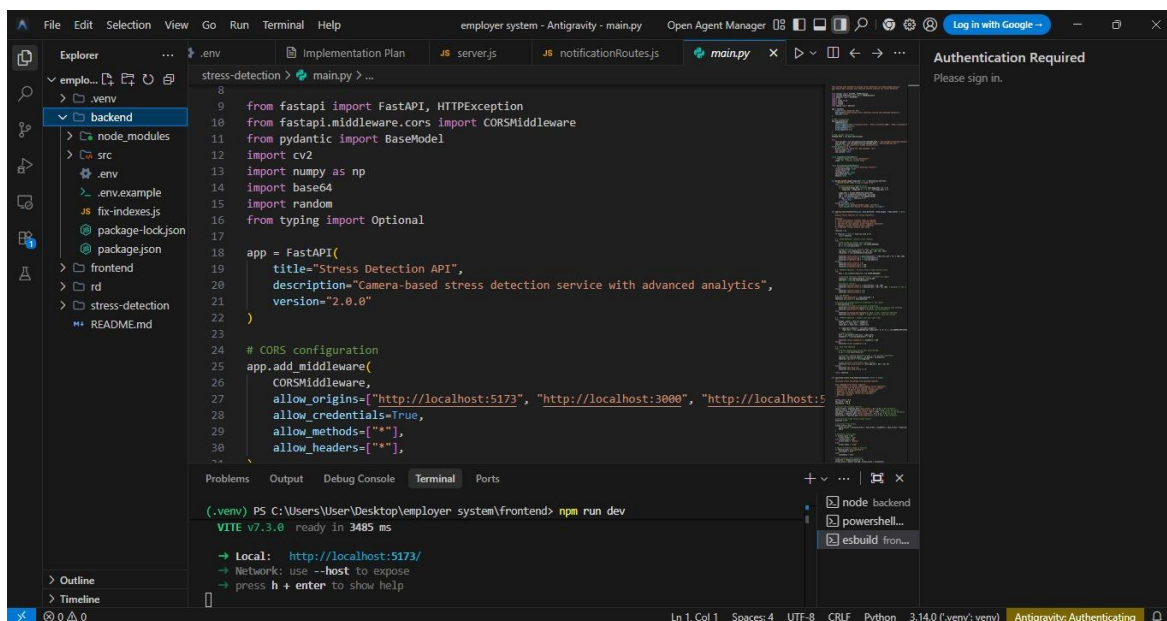
- Admin and Employee dashboards provide clear visibility into stress and workload information.
- UI workflows are intuitive and align with defined requirements.
- Role separation improves data privacy and system security.

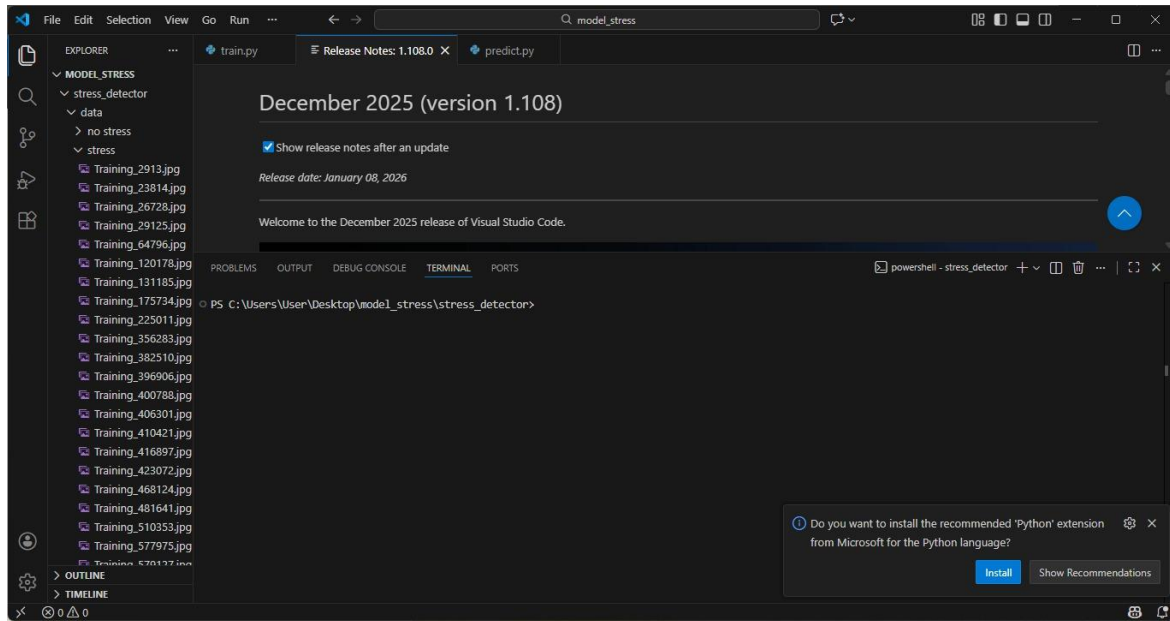
4. Overall Component-Level Findings

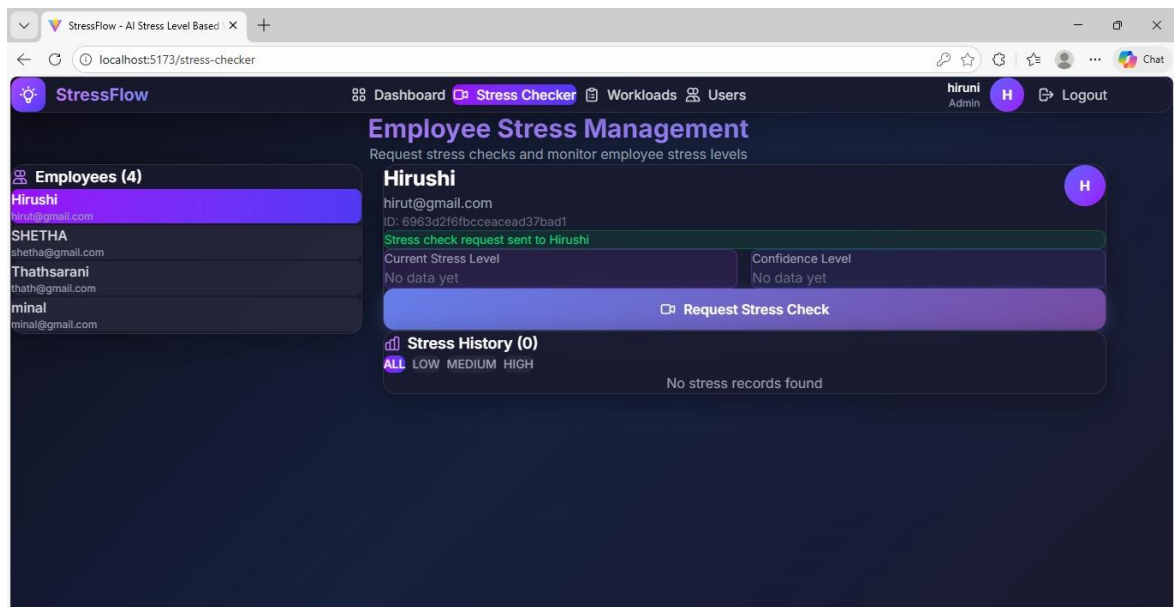
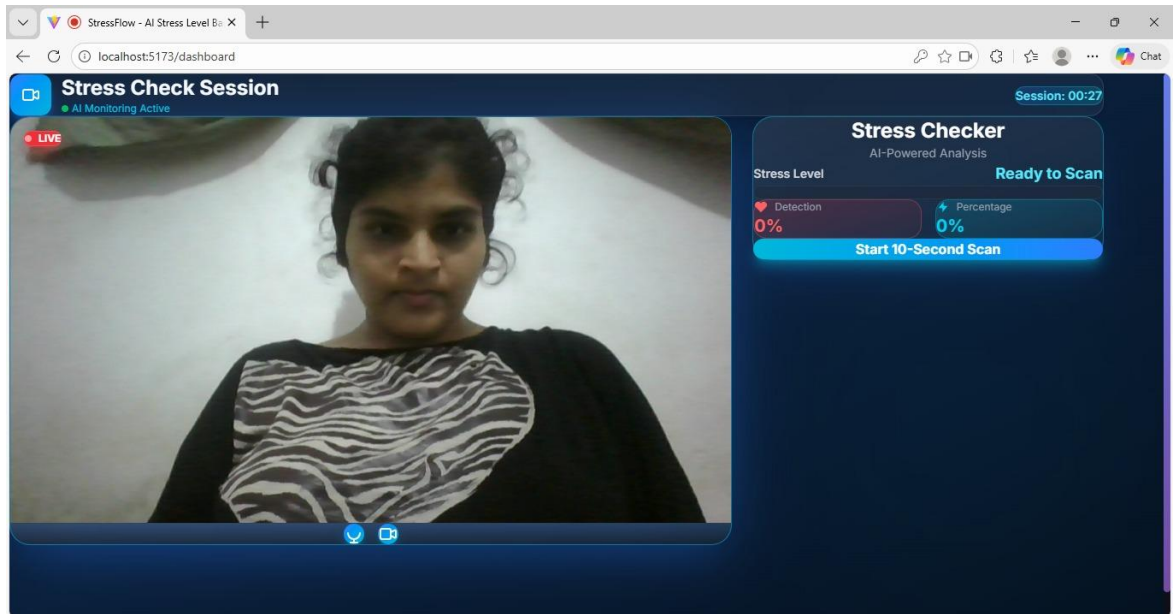
- The component successfully demonstrates the feasibility of integrating **AI-based stress detection with task and workload management**.
- User interfaces support both monitoring and decision-making without overwhelming users.
- The component complements Component 01 by providing the **human-facing and operational control layer** of the system.
- Current results validate the design and implementation approach at the PP1 stage.

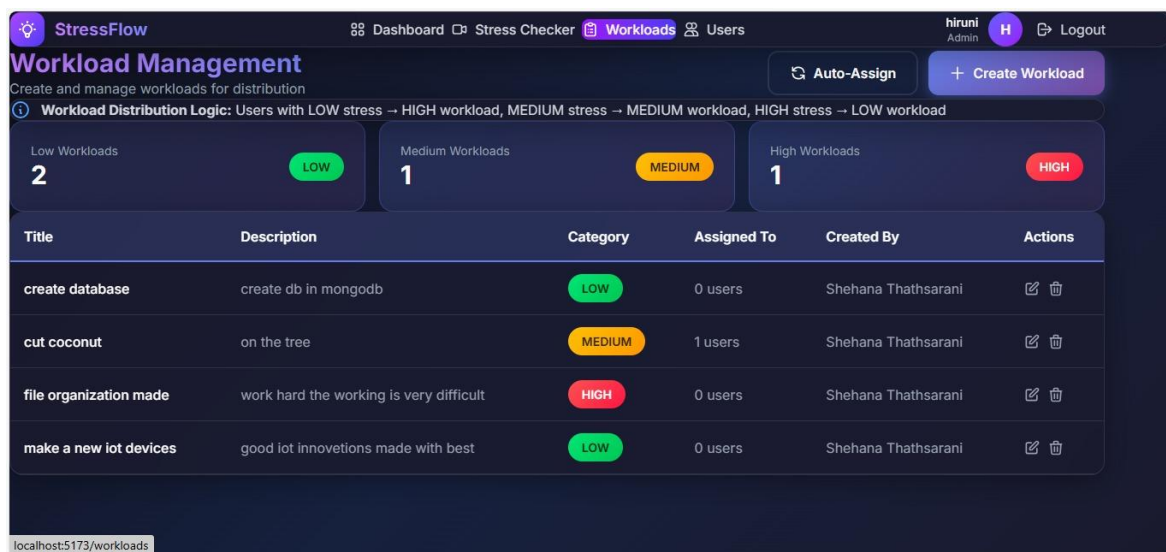
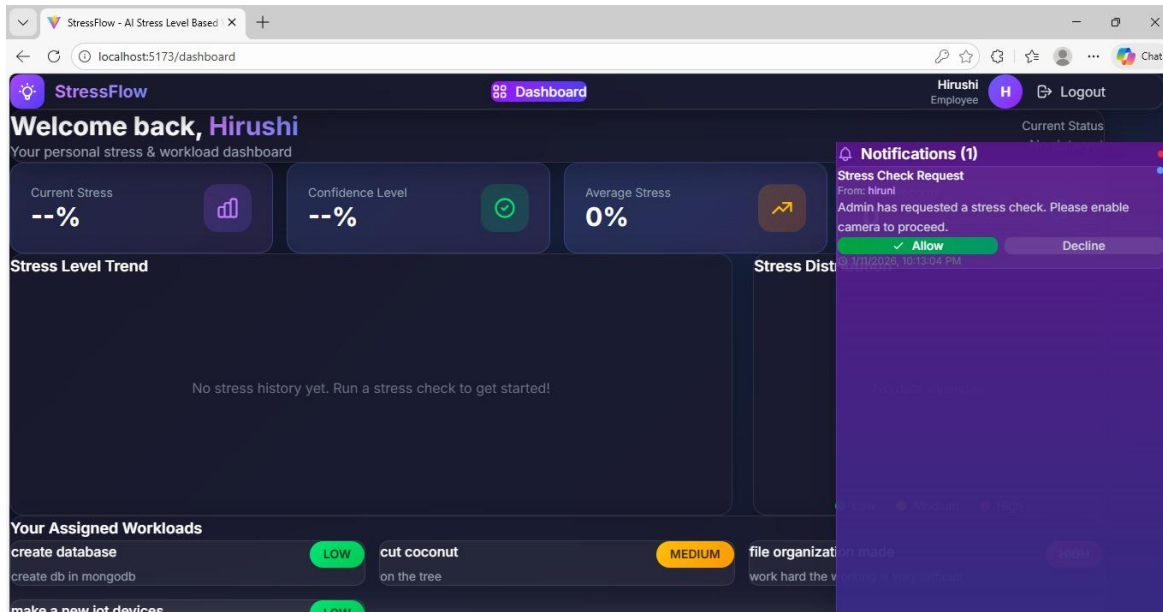
Current Limitations and Future Improvements

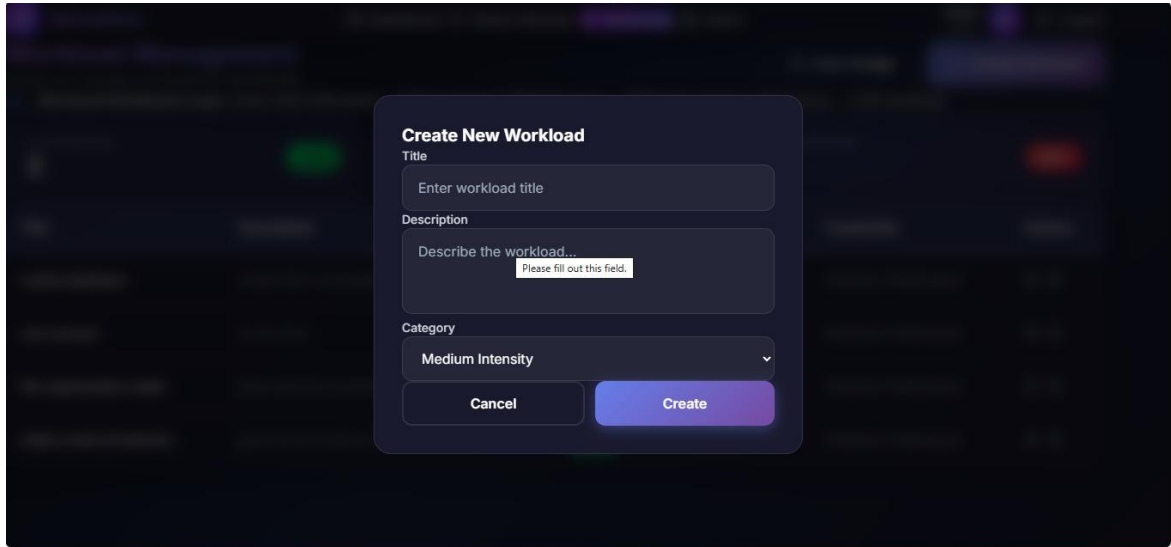
- Stress detection accuracy may vary due to lighting and facial variations.
- Larger datasets and improved feature extraction are planned.
- Deeper integration with automated workload reallocation logic will be implemented in future phases.











Methodology

The proposed methodology follows a layered and modular approach to design an AI-driven smart office system that integrates stress detection, workload analysis, intelligent decision-making, and visualization. The methodology is divided into multiple interconnected components to ensure scalability, reliability, and adaptability.

The first methodological component focuses on workload and task analysis using machine learning. Based on findings in the literature, different office activities such as video conferencing, software development, data analysis, and document handling exhibit distinct workload and resource usage patterns. To capture these patterns, structured datasets were prepared using task-related features and activity indicators. Feature selection was applied to retain only the most relevant attributes, reducing noise and improving model interpretability. Multiple supervised machine learning algorithms, including Logistic Regression, Random Forest, and Neural Networks, were implemented and compared. These models form the basis for task type classification (Model 01) and workload intensity classification (Model 02). Standard classification metrics such as accuracy, F1-score, AUC, and confusion matrices were used to evaluate baseline performance.

The second methodological component addresses resource and bandwidth usage prediction. Regression-based machine learning techniques were used to estimate expected system or network resource demand based on task type, workload intensity, and historical usage patterns. This resource prediction model (Model 03) enables proactive planning rather than reactive system adjustments. Although real-time enforcement on physical network devices is not implemented at this stage, the model is designed to support future

integration with SD-WAN or QoS controllers. Model performance was evaluated using error-based metrics such as Mean Squared Error (MSE) and Root Mean Squared Error (RMSE), along with trend comparisons.

The third methodological component focuses on AI-based stress detection using computer vision. Facial features extracted from camera input are processed using image preprocessing techniques such as resizing, normalization, and noise reduction. A machine learning classifier predicts employee stress levels under controlled conditions. Stress data is timestamped and stored for trend analysis, supporting both real-time monitoring and historical evaluation. This component enables non-intrusive stress monitoring and forms a key input for workload-aware decision-making.

The fourth methodological component introduces a feedback-based learning and optimization mechanism (Model 04). System feedback, including previous predictions, workload outcomes, and performance metrics, is used to support incremental model refinement. This feedback loop allows the system to adapt to changing user behavior and workload patterns over time, forming a closed-loop intelligent framework.

The final methodological component focuses on data storage, integration, and visualization. All predictions, stress records, workload data, and logs are stored in MongoDB due to its flexibility in handling heterogeneous and time-stamped data. A web-based dashboard provides real-time visualization of stress levels, workload distribution, and task status. This layered methodology ensures seamless integration of analytics, decision support, and user interaction into a unified smart office platform.

Challenges and Overcome Planes

One of the major challenges encountered during the project was the availability and quality of datasets for both workload analysis and stress detection. Real-world office datasets that include labeled task types, workload intensity, and stress levels are limited due to privacy, ethical, and organizational constraints. This challenge was addressed by generating controlled and synthetic datasets based on realistic office scenarios and usage patterns reported in existing studies. This approach enabled balanced class representation and supported the early training and validation of machine learning models.

Training and validating four interrelated machine learning models introduced additional complexity. Each model serves a distinct purpose—task classification, workload intensity estimation, resource prediction, and feedback-based learning—requiring careful feature selection and data consistency across models. During early experiments, variations in feature distributions and class imbalance affected model performance. These issues were mitigated through feature normalization, class balancing, and algorithm comparison. Multiple algorithms, including Logistic Regression, Random Forest, and Neural Networks, were evaluated to identify suitable baseline models for each task.

Another significant challenge involved model performance stability and overfitting, particularly for workload classification and regression models trained on limited datasets. To overcome this, baseline models were selected first, and performance was evaluated using multiple metrics such as Accuracy, F1-score, AUC, confusion matrices, and error-based regression metrics rather than relying on a single indicator. This ensured more reliable assessment of model behavior at the PP1 stage.

Stress detection presented its own challenges due to variability in facial expressions, lighting conditions, and camera positioning. These factors affected the consistency of predictions during early testing. Image preprocessing techniques such as resizing, normalization, and noise reduction were applied to improve robustness. In addition, stress predictions were designed to be supportive indicators rather than definitive assessments, reducing the risk of incorrect interpretation.

From a system integration perspective, combining machine learning models, stress detection modules, dashboards, and database storage increased overall system complexity. This was addressed by adopting a modular architecture, where each component operates independently with clearly defined interfaces. MongoDB was selected to manage heterogeneous data such as predictions, timestamps, and logs efficiently.

Privacy and ethical considerations also posed challenges, as stress and workload data are sensitive. This was addressed through role-based access control, restricted visibility of individual stress records, and secure data handling practices. Ethical disclaimers were incorporated to clarify that the system does not provide medical diagnoses.

Conclusion

This project presents the ongoing development of an **AI-driven smart office system** designed to monitor employee stress, analyze workloads, and support intelligent task and resource management. At the current stage, approximately **50% of the system has been designed and implemented**, consistent with the PP1 progress presentation.

Key components developed so far include **four machine learning models for task classification, workload intensity estimation, resource prediction, and feedback-based learning**, along with an AI-based stress detection module and interactive dashboards. Preliminary evaluation results demonstrate that the selected features, models, and system architecture are suitable for intelligent workload and stress analysis under controlled conditions.

However, several aspects of the system remain under active development, including full backend integration, automation of decision logic, extended testing, and optimization of model performance. Future work will focus on improving prediction accuracy, enhancing

system integration, validating the system under more realistic conditions, and preparing the solution for final evaluation.

Overall, the work completed to date establishes a **solid technical foundation** for the proposed smart office platform. With continued development and validation, the system shows strong potential to evolve into a scalable, intelligent solution that supports employee wellbeing and efficient workload management in modern office environments.